An Algorithm for Indoor Lighting Layout Optimization Using Simulated Annealing

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Abstract

Indoor lighting is a major contributor to building energy consumption, and optimizing its layout is essential for reducing energy use while maintaining visual comfort. However, most existing studies focus on improving luminaire efficiency or daylighting design, while systematic research on luminaire layout optimization remains limited. This paper proposes an optimization method for indoor lighting layout based on the Simulated Annealing (SA) algorithm, aiming to achieve an energy-efficient lighting configuration through intelligent optimization. The method comprehensively considers lighting quality and energy consumption by constructing a multi-objective function, which is then solved iteratively using SA. A case study demonstrates that this method can effectively reduce lighting energy consumption while ensuring indoor lighting quality, offering a new approach and tool for indoor lighting design.

Keywords: Simulated Annealing; Lighting Optimization; Energy Consumption; Layout Algorithm

1 Introduction

In modern society, people spend approximately 87% of their time indoors[1], and most human information is acquired through vision stimulated by light. Consequently, indoor lighting has a significant impact on human health and productivity. A well-designed lighting environment supports circadian rhythms and visual comfort, while poor lighting conditions may lead to sleep disorders, endocrine imbalances, and reduced efficiency[2].

Furthermore, the energy consumption generated by indoor lighting cannot be overlooked. As the second largest energy-consuming system after air conditioning [4], lighting accounts for approximately 20% of a building's electricity use[3].It can thus be concluded that contemporary lighting design must not only create a scientific luminous environment but also rationally plan energy consumption to meet the development principles and objectives of green energy conservation[4].

Previous research has primarily focused on three approaches: (1) developing more efficient luminaires such as LEDs; (2) enhancing daylighting utilization; and (3) optimizing luminaire placement. The first two approaches face diminishing returns or environmental limitations[5], making luminaire layout optimization the most stable and effective strategy[6]. Studies have shown that rational layout planning can reduce energy consumption by up to 25% [7]. However, most existing methods rely on heuristic design rather than intelligent optimization, which limits their scalability.

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Lighting analysis[8] is typically conducted using simulation tools such as DIALux and Radiance [9]. These tools provide accurate evaluations, but the optimization of layouts remains an NP-hard problem that requires advanced algorithms. Among intelligent optimization methods, Genetic Algorithms (GA), Particle Swarm Optimization (PSO), and Simulated Annealing (SA) are widely used. SA is particularly attractive due to its global search capability and robustness.

To address the challenges outlined above, this paper proposes a lighting layout optimization method based on the Simulated Annealing (SA) algorithm. The method minimizes energy consumption while ensuring visual comfort by optimizing luminaire positions and quantities. The evaluation framework in this study references multiple standard systems, including both international and national lighting codes. Experimental results demonstrate its effectiveness in achieving a balanced and efficient lighting configuration.

The remainder of this paper is organized as follows: Section 2 introduces the overall framework, Section 3 details the methodology, Section 4 presents the case study and results, Section 5 provides the discussion, and Section 6 concludes the work.

2 Basic Idea

This paper proposes an indoor lighting layout optimization method based on the Simulated Annealing algorithm. The core idea is to optimize the positions and quantities of luminaires using SA to minimize energy consumption while ensuring lighting quality. The initial CAD format drawings are converted into IFC format files containing luminaire data. Subsequently, lighting calculations are performed using DIALux and Radiance, followed by spatial evaluation. Based on these results, an objective function is established and solved through simulated annealing. A flowchart of the algorithm is presented in Figure 1.

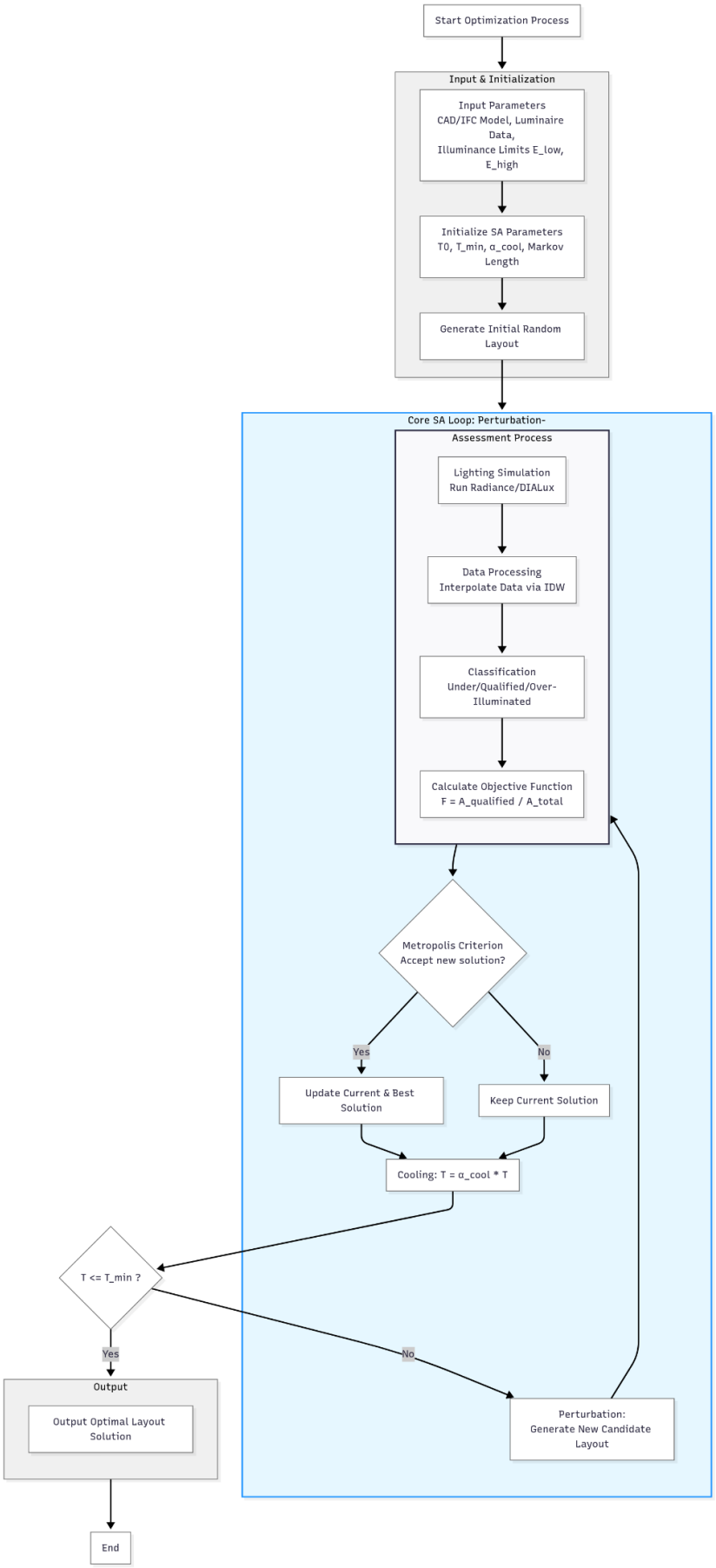


Figure 1: Flowchart of the Simulated Annealing Optimization Process

3 Methodology

The Simulated Annealing (SA) algorithm is a stochastic global optimization technique inspired by the annealing process in metallurgy[10]. It accepts not only superior solutions but also, with a certain probability, inferior ones to escape local optima. In the field of performance optimization for other building facility layouts, the use of genetic algorithms such as Simulated Annealing has proven to be quite effective[11]. In this study, SA is adapted to the luminaire layout problem by defining an objective function that incorporates both lighting quality and energy consumption.

3.1 Assessment method

The quality of indoor lighting depends on both the performance and placement of luminaires. A room with an average illuminance that meets standards may still contain localized areas that are excessively bright or dim due to poor arrangement. Although the national standard GB/T 50034 provides detailed specifications for evaluating lighting quality—such as task plane illuminance, immediate surrounding illuminance, and uniformity—in practice, designers and reviewers often focus only on whether the average illuminance meets the minimum requirement. This may lead to problems such as insufficient illuminance in non-task areas, localized over-illumination on task surfaces, or overall excessive lighting, which can reduce visual comfort and waste energy.

To provide a more detailed and intuitive reflection of indoor lighting quality, this study establishes upper and lower limits for illuminance based on design specifications for task plane illuminance, surrounding illuminance, and lighting power density (LPD). These limits are used to create grayscale illuminance maps: areas below the lower limit are classified as non-compliant, those between the limits as compliant, and those above the upper limit as excessive.

The upper and lower limits of the qualified task plane illuminance are calculated as follows:

- Lower illuminance limit: Standard illuminance value × 0.9.
- Upper illuminance limit: Derived from the LPD limit, assuming a luminaire efficacy of 100 lm/W, using equations (1) and (2):

$$E_{av} = \frac{N\varphi UK}{A} \tag{1}$$

$$LPD = \frac{N\varphi}{100A} \tag{2}$$

where E_{av} is the average illuminance on the task plane, ϕ is the luminous flux per luminaire, N is the number of luminaires, U is the utilization factor (typically 0.7), K is the maintenance factor (typically 0.8), and A is the area of the task plane.

3.2 Objective Function Construction

The objective function comprehensively accounts for both lighting energy consumption

and quality metrics. It is defined as:

$$F = A_{\text{qualified}}/A_{\text{total}}$$

where:

- $A_{\text{qualified}}$ is the area where illuminance falls within the qualified range.
- A_{total} is the total area of the given space.

3.3 Perturbation Mechanism

Within SA, new candidate solutions are generated by perturbing the current solution. Perturbation operations include adjusting the position of one or more luminaires. This strategy ensures both local exploration and global search capability, balancing solution diversity with convergence efficiency.

3.4 Iteration Process

The iterative process of the algorithm consists of the following steps[12]:

1. **Initialize parameters**: Set the initial temperature T_0, final temperature T_{end}, cooling schedule coefficient α_{cool}, and Markov chain length.
2. **Generate initial solution**: Create a random initial luminaire layout configuration.
3. **Evaluate current solution**: Calculate the objective function value F for the current layout.
4. **Generate new candidate solution**: Apply the perturbation mechanism to the current solution to create a new layout and calculate its objective function value F'.
5. **Acceptance criterion**: Decide whether to accept the new solution based on the Metropolis criterion (accept if $F' < F$, or with probability $\exp(-(F' - F)/T)$ if $F' \geq F$).
6. **Cooling**: Reduce the temperature according to the cooling schedule (e.g., $T_{k+1} = \alpha_{cool} \cdot T_k$).
7. **Termination check**: If the temperature has reached the final temperature T_{end}, output the best solution found; otherwise, return to Step 4.

4 Case Study

4.1 Experimental Design

The experimental subject is a residential apartment with a typical layout. To optimize the lighting design, the SA algorithm was applied. Minor perturbations were introduced to the initial positions of the luminaires within a constrained range to generate new candidate solutions. The algorithm parameters were configured based on the spatial characteristics

of the layout: the inner loop was set to 20 iterations, the cooling rate was set to 0.6, and five major cycles were implemented to avoid convergence to local optima.

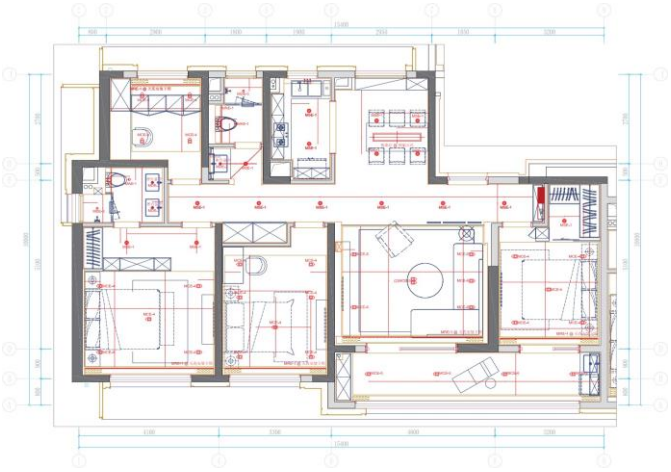


Figure 2: Apartment layout

4.1.1 Illuminance Data

The point-wise illuminance data for the experimental area were first calculated by Radiance using a ray-tracing algorithm and then interpolated to a finer resolution through inverse distance weighting (IDW) in ArcGIS, as shown in Figure 3. The data were then reclassified using the method described in Section 3.1: yellow indicates areas exceeding the upper limit, blue indicates areas below the lower limit, and green represents compliant areas.

As shown in Figure 5b, the compliant green zone is relatively limited, while the yellow and blue areas representing over- and under-illuminance are disproportionately large. Based on this observation, the SA algorithm was employed to perturb luminaire positions and recalculate the compliance rate in pursuit of improved performance, using one representative room as a case study.

4.2 Experimental Results

During the optimization process, the compliance rate improved substantially, increasing from approximately 63% in the original layout to over 80%, as shown in Figure 4. The specific changes in luminaire arrangement and the resulting illuminance distribution are presented in Figure 5.



(a) Interpolated data (b) Interpolated data (classified)

Figure 3: Illuminance data distribution

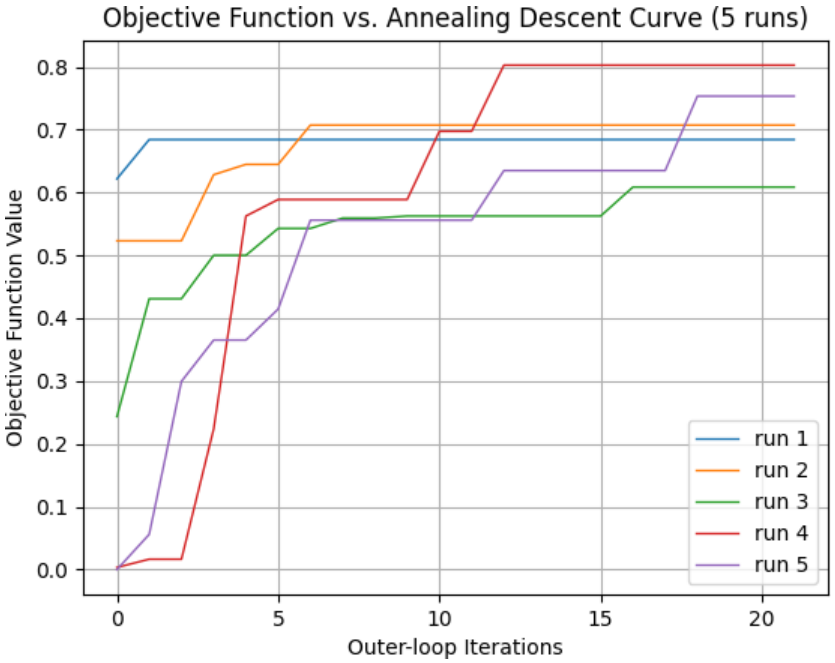


Figure 4: Variation of the objective function during optimization

5 Discussion

The integration of the Simulated Annealing (SA) algorithm with the proposed assessment method provides a robust framework for optimizing lighting layouts. The compliance rate increased from about 63% to over 80%, demonstrating that SA can effectively manage the complex search space of luminaire placement. It identifies configurations that enhance lighting quality while reducing energy consumption.

5.1 Optimization Mechanism and Parameter Selection

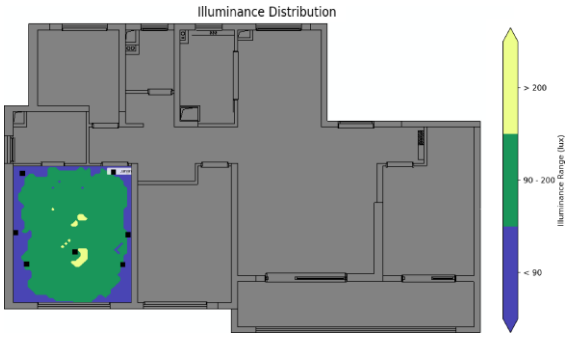
The iteration count of 20 offered a good balance between computational efficiency and solution quality. This setting ensured adequate exploration of the solution neighborhood at each temperature stage without excessive computation. The cooling rate of 0.6 facilitated rapid convergence given the size of the experimental space. In addition, the use of five major cycles prevented the algorithm from being trapped in local optima.

5.2 Synergy with Assessment Methodology

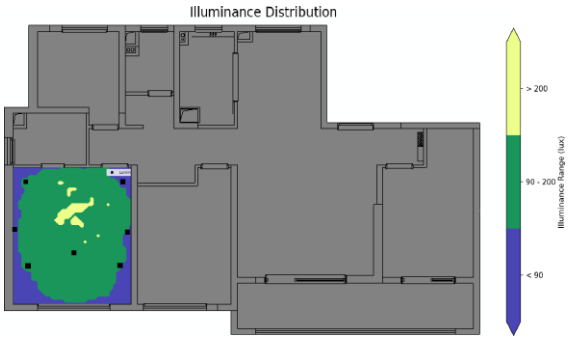
The SA algorithm worked effectively with the quantitative metrics provided by the assessment framework. The classification of areas into non-compliant, compliant, and excessive created clear optimization targets. Each perturbation in luminaire placement generated measurable changes in these categories, providing immediate feedback and guiding the optimization process. This integration enabled decisions that simultaneously addressed multiple performance criteria: reducing energy waste in over-lit areas and improving visual comfort in under-lit areas.



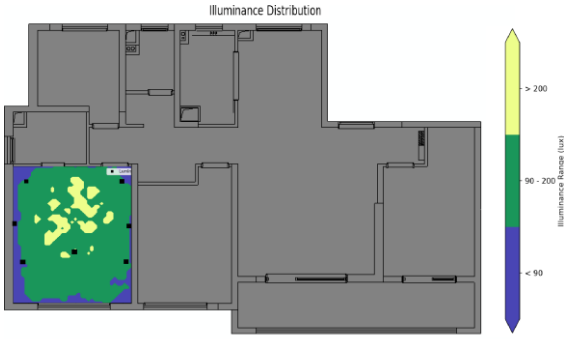
(a) Step 1



(b) Step 2



(c) Step 3



(d) Step 4

Figure 5: Evolution of luminaire layout during optimization

5.3 Practical Implications and Adaptability

The proposed framework demonstrates practical value because it achieves improvements in both energy efficiency and lighting comfort. The method is adaptable to various spatial configurations and lighting requirements, suggesting broad applicability across different building types and use scenarios.

Overall, this approach represents a significant advancement over conventional lighting design methods, offering a systematic and performance-driven methodology that effectively balances multiple competing objectives through intelligent optimization.

6 Conclusion

This research presents a robust and versatile methodology for indoor lighting optimization using multi-cycle simulated annealing. The framework is not limited to specific layouts or lamp types. Instead, it can be adapted to a variety of residential and commercial spaces by adjusting key parameters such as optimization bounds and the number of iterations, making it a practical tool for architects and lighting designers.

The significance of this work is twofold. First, from a human-centric perspective, it contributes to enhancing living environments by enabling the design of visually comfortable and uniform lighting conditions, which are crucial for occupant well-being and productivity. Second, from a sustainability perspective, the method achieves these improvements without compromising energy efficiency. By minimizing over-lighting and reducing energy waste, the optimized designs directly lower electricity consumption and, consequently, the carbon footprint of buildings. This aligns with global sustainability goals and provides a pathway toward healthier, more comfortable, and more energy-efficient built environments.

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